SEC P vs NP log 2026-05-12

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 10:00 UTC

SEC P vs NP — daily log 2026-05-12

23 cycles recorded

Free Cumulant Sum Bounded by Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 01:18 UTC
- **Entry ID**: `ecd8f062c47d`

Statement

For a |X|×|Y| matrix M defining a partial function f, let μ(M) = ∑_{k≥1} |c_k(M)| where c_k(M) are the free cumulants of M's entries. Then μ(M) ≤ O(√(R(f))) where R(f) is the randomized communication complexity of f. For DISJ_n, μ(DISJ_n) = Ω(n) holds.

Rationale

Free cumulants capture non-commutative independence structures in matrix entries, which may correlate with the information-theoretic cost of coordinating disjointness checks. The sum's growth rate could reflect the minimal communication required to resolve entangled dependencies.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 54, in run_trial
    cumulants = free_cumulants(M)
                ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 39, in free_cumulant
   R[i][j] -= sum(R[x][y] * R[i - x][j - y] for x in range(i) for y in range(j)) / (i * j)
            ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed due to division by zero, preventing evaluation of conjecture | next: Fix implementation to handle zero division and retest with validated inputs

---

Plethysm Coefficient Gap in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Size for Permanent Polynomial
- **Recorded**: 2026-05-12 01:34 UTC
- **Entry ID**: `40591186a3ea`

Statement

For all $n \geq 2$, the plethysm coefficient λ(perm_n) of the symmetric power S^k(perm_n) exceeds λ(det_m) + $\varepsilon \cdot n^2$ for m < n^{1.5}, where $\varepsilon > 0$ is universal. This gap persists under linear substitutions preserving the polynomial's symmetry type.

Rationale

Plethysm coefficients encode symmetry constraints of polynomial representations. By linking them to circuit complexity, we may expose algebraic barriers to expressing permanent as a determinant-like circuit, leveraging representation-theoretic invariants resistant to natural proofs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.19s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 58, in run_trial
    if lambda_perm_n > lambda_det_m + epsilon * n ** 2:
                       ~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~
OverflowError: int too large to convert to float
```

Judge reasoning

Test crashed with overflow error before producing data; cannot verify conjecture's validity | next: Implement arbitrary-precision arithmetic or numerical stability checks for large n

---

Symmetric Group Fourier Coefficients Inverse Proportional to ACC⁰ Circuit Size for Sipser Function

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Finite Groups × ACC^0 Circuit Size for Sipser Function
- **Recorded**: 2026-05-12 01:48 UTC
- **Entry ID**: `567baddf9ba5`

Statement

For the Sipser function S_n on n variables, the maximum absolute Fourier coefficient over the symmetric group S_n's irreducible representations is $\Theta(1/k)$, where k is the minimal ACC^0 circuit size computing S_n. This holds for all $n \geq 2$.

Rationale

Symmetric group representations capture combinatorial symmetries of permutation-based functions like Sipser's. Fourier coefficients over these representations quantify 'non-abelian harmonic analysis' of the function, potentially revealing structural obstructions to efficient ACC^0 computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 59, in run_trial
    for k in range(1, 2**n + 1):
                         ^
NameError: name 'n' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'n' before producing results | next: Fix the NameError in test code by defining 'n' before usage

---

Integer Point Count in Solution Polytope Bounds Extended Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometry of Numbers × Extended Frege Proof Length
- **Recorded**: 2026-05-12 01:54 UTC
- **Entry ID**: `4798bdb0751b`

Statement

For a 3-SAT instance Φ with n variables, the number of integer solutions S(Φ) satisfies S(Φ) ≥ 2^{n - L(Φ)}, where L(Φ) is the length of the Extended Frege proof of Φ. Equality holds when Φ is a tautology with a unique solution.

Rationale

The geometry of numbers connects the combinatorial structure of integer solutions to algebraic constraints, while Extended Frege proofs formalize logical deductions. The conjecture posits that solution density directly impacts proof complexity via exponential trade-offs, exposing hidden geometric regularities in tautology refutations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in run_trial
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in <genexpr>
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 27, in dpll
    literal, rest = sat_formula[0]
    ^^^^^^^^^^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

Test crashed during execution, preventing data collection. The error suggests a structural issue with the sat_formula handling in the DPLL algorithm. | next: Debug the DPLL implementation's handling of sat_formula structure to resolve the unpacking error

---

Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Nisan-Wigderson PRG Seed Length
- **Recorded**: 2026-05-12 02:14 UTC
- **Entry ID**: `fa70962e79f5`

Statement

For any 3-SAT instance with n variables, the seed length of the Nisan-Wigderson PRG fooling it is at most the rank of the matroid formed by its clause hypergraph, where the rank is the maximal size of a set of clauses with pairwise disjoint variable support.

Rationale

Matroid rank captures combinatorial dependencies between clauses, which may constrain the pseudorandomness needed to fool the formula. This bridges structural properties of CNF formulas with PRG efficiency, leveraging matroid theory's role in encoding combinatorial constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and memory limits to complete execution

---

Matroid Rank Submodularity and Monotone Circuit Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 02:20 UTC
- **Entry ID**: `b052cc185466`

Statement

For a k-CLIQUE CNF formula Φ on n variables, the rank function r(S) of the matroid M constructed from Φ satisfies r(S) ≥ Ω(n^{1/2}) for any subset S of size Ω(n^{1/2}), and the minimum number of elements needed to achieve r(S) = Ω(n^{1/2}) is Ω(n^{1/2}).

Rationale

The submodularity of the rank function in matroid theory could mirror the structure of monotone circuits, where dependencies between clauses (elements) affect the circuit's size. By linking the rank's growth rate to the circuit's complexity, we might uncover structural constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with detailed logging to identify crash cause and collect empirical evidence

---

Symplectic Rank of Disjointness Communication Matrix Bounds Randomized Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symplectic Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 02:46 UTC
- **Entry ID**: `8acf355ada09`

Statement

The symplectic rank of the DISJOINTNESS communication matrix M is Ω(n), where the symplectic rank is defined as the minimal k such that M decomposes into k rank-1 symplectic tensors under a fixed symplectic form. This lower bounds the randomized communication complexity of DISJOINTNESS.

Rationale

Symplectic geometry provides invariants that capture structural properties of matrices, which may correlate with communication complexity. The symplectic rank, as a decomposition invariant, could expose hidden algebraic constraints in the matrix that mirror information-theoretic barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 74, in run_trial
    if not is_symplectic_form(M, n):
           ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 44, in is_symplectic
    return matrix_multiply(matrix_multiply(M, M), I) == -I and matrix_multiply(transpose(M), M) == -
I
                                                                                                  ^^
TypeError: bad operand type for unary -: 'list'
```

Judge reasoning

Test crashed due to type error in matrix operations, preventing evaluation of conjecture | next: Fix matrix arithmetic implementation to handle symplectic form verification correctly

---

Schur Coefficient Density Inverse Proportional to Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric Function Theory × Frege Proof Length
- **Recorded**: 2026-05-12 03:10 UTC
- **Entry ID**: `2b3f03ad1b52`

Statement

For a homogeneous polynomial f of degree d in n variables, let S(f) denote the number of non-zero coefficients in its Schur decomposition. Then, the Frege proof length of f = 0 is at least Ω(S(f)^{1/2})

Rationale

The density of Schur coefficients reflects the complexity of the polynomial's representation, which might correlate with the difficulty of proving its identity through Frege systems. Symmetric function theory provides a framework to decompose polynomials into irreducible components, exposing structural properties that could influence proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
ue': 1, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                                   ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed due to KeyError before producing reliable data | next: Fix the test code to handle missing 'seed' field and re-run with proper error handling

---

Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 03:25 UTC
- **Entry ID**: `93b47d86db79`

Statement

For any read-twice BP P computing IP_2, the sum of free cumulants Σ_{k=1}^∞ κ_k(P) ≤ log(size(P)) + O(1). For random read-twice BPs over GF(2), Σ κ_k(P) = Θ(log n).

Rationale

Free cumulants capture non-commutative independence structures; their sum could distinguish read-twice BPs (with limited independence) from unrestricted BPs (with full independence). IP_2's inherent symmetry might force linear cumulant growth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing supporting data or counterexamples | next: Increase timeout duration and re-run tests with extended computation limits

---

Condition Number of Moment Matrix Inversely Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex Geometry (Condition Number of Moment Matrices) × SOS Degree for Max-CUT
- **Recorded**: 2026-05-12 04:34 UTC
- **Entry ID**: `f89319bf92ad`

Statement

For a random max-CUT instance with n variables, the condition number κ of the moment matrix in the degree-d SOS relaxation satisfies $\kappa = \Theta(1/d^2)$, and the SOS degree required to achieve an approximation ratio of 0.878 - ε is $\Omega(\sqrt{(d)})$

Rationale

The condition number of the moment matrix reflects the sensitivity of the SOS relaxation to perturbations. A lower condition number implies better-conditioned matrices, which may require lower SOS degrees to achieve the same approximation ratio. This link could reveal geometric properties of the feasible region in the SOS hierarchy

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 65, in run_trial
    inv_M = matrix_inv(M)
            ^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 42, in matrix_inv
    A_rref = gaussian_elimination(A_augmented)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 28, in gaussian_elim
    raise ValueError("Matrix is singular")
ValueError: Matrix is singular
```

Judge reasoning

Test crashed with singular matrix error, preventing validation of conjecture | next: Investigate why matrix is singular for tested parameters and verify matrix construction

---

Projective Plane Incidence Bounds Nisan-Wigderson Seed Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Geometry (Projective Planes) × Nisan-Wigderson PRG Seed Length
- **Recorded**: 2026-05-12 04:40 UTC
- **Entry ID**: `c9054923e964`

Statement

For a CNF Φ with n variables whose clause incidence matrix is a projective plane of order q, the Nisan-Wigderson seed length for Φ is $\leq q^2 + q + 1$. Conversely, if Φ has seed length $> q^2 + q + 1$, its incidence matrix cannot be a projective plane of order q.

Rationale

Projective planes exhibit optimal combinatorial expansion properties that could constrain pseudo-random generator seed requirements. The bound q^2+q+1 matches the number of lines in the plane, suggesting a direct geometric constraint on the PRG's pseudorandomness threshold.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 148, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 110, in run_trial
    cnf = generate_cnf_from_plane(points, lines)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 85, in generate_cnf_
    clause = [j + 1 for j in range(m) if (i // q, i % q) in lines[j]]
                                               ^

NameError: name 'q' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'q' in the generation logic, preventing validation of the conjecture | next: Fix the code to properly define 'q' in the projective plane generation and re-run tests

---

Fourier Coefficient Spread and ACC^0 Circuit Lower Bounds

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Finite Groups × ACC^0 Circuit Size
- **Recorded**: 2026-05-12 05:17 UTC
- **Entry ID**: `90abe309451a`

Statement

For any boolean function $f:\{0,1\}^n \to \{0,1\}$, if the sum of absolute Fourier coefficients $\sum|\hat{f}(S)| \geq 2^{n/2} \sqrt{n}$, then f cannot be computed by an ACC^0 circuit of size $2^{n/2}$.

Rationale

The spread of Fourier coefficients reflects correlation with parity functions. ACC^0 circuits, limited to bounded-depth polynomial-size gates, struggle to capture such correlations, making large Fourier sums a barrier to efficient computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.07s`

```
tric_name': 'Fourier Coefficient Sum', 'metric_value': 0.64453125, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.626953125, 'instances_tested': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5625, 'instances_tested': 1, 'con
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.634765625, 'instances_tested': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.546875, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.56640625, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5546875, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.658203125, 'instances_tested': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.666015625, 'instances_tested': 1
RESULT: SUPPORTED mean=0.6087239583333334 std=0.03415580763030604 support_fraction=1.0
```

Judge reasoning

The test used n=1 where the conjecture's threshold ($2^{n/2}\sqrt{n}$) is ~1.414, but all metric values (<1) never reached this threshold. Thus, the conjecture's condition was never tested. | next: Test with n≥2 where $2^{n/2}\sqrt{n} \geq 2$ to validate the conjecture's actual condition

---

Fourier Coefficient Decay Bounds AC⁰ Circuit Size for Parity-Insensitive Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Finite Groups × AC⁰ Circuit Size
- **Recorded**: 2026-05-12 05:43 UTC
- **Entry ID**: `29d6f7d21e9f`

Statement

For any parity-insensitive Boolean function f: {0,1}^n → {0,1}, the sum of absolute Fourier coefficients ∑_{S≠∅} |f̂(S)| decays exponentially with n iff the minimal AC⁰ circuit size computing f is Ω(2^{n/2}).

Rationale

Parity-insensitive functions exhibit Fourier coefficient concentration patterns that may correlate with algebraic complexity. The exponential decay threshold could expose a structural gap between Fourier analytic simplicity and circuit complexity, leveraging the inherent symmetry of AC⁰ computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.29s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 67, in run_trial
    f_hat = compute_fourier_coefficients(clauses, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 38, in compute_fouri
    return [abs(coeff / (1 << n)) for coeff in fast_walsh_hadamard_transform(f)]
                                               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 23, in fast_walsh_ha
    f[j] += f[j + n // 2]
            ~^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity | next: Debug fast_walsh_hadamard_transform implementation to fix index out of range error

---

Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Invariant Theory × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-12 06:36 UTC

- **Entry ID**: `3a288632d238`

Statement

For a symmetric CSP instance Φ invariant under a finite group G, the minimal degree d of a generating set of the invariant ring of G is inversely proportional to the SOS refutation degree of Φ. Specifically, the SOS refutation degree is at least $\Omega(1/d)$.

Rationale

Symmetric CSPs can be analyzed via their invariant rings, which capture the group's action. The degree of the generators may limit the SOS hierarchy's ability to refute the instance, as higher-degree generators might require higher SOS levels.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 51, in run_trial
    instance = generate_symmetric_csp(3)  # Example: n=3
               ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 37, in generate_symm
    while not is_symmetric(instance):
              ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 30, in is_symmetric
    permuted_instance = [instance[g[i]] for i in range(len(instance))]
                                  ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity. | next: Fix the index out-of-range error in generate_symmetric_csp() and re-run tests with multiple seeds

---

Noncommutative L^p Norm of Communication Matrix Bounds Randomized Disjointness Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 06:43 UTC
- **Entry ID**: `cfd6712a7d9a`

Statement

For a random instance M of DISJOINTNESS on n elements, the noncommutative L^p norm ||M||_{p} ≥ Ω(√n) for p=2, with equality achieved when M is a balanced bipartite graph with uniform edge weights.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of tensors, which may encode entanglement-like constraints in communication protocols. The L^2 norm's relation to matrix rank could expose hidden geometric barriers in randomized protocols for DISJOINTNESS.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.10s`

```
c_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30, 'conjecture_h
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
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TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
RESULT: SUPPORTED mean=28.284271247461902 std=undefined support_fraction=1.0
```

Judge reasoning

Safety rail: critic_challenge | original judge: | next: NONE

Matroid Rank Inverse Proportional to Monotone DNF Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 07:48 UTC
- **Entry ID**: `afe153586854`

Statement

For a k-CLIQUE instance on n vertices, let r be the maximum number of pairwise disjoint k-cliques. Then, the minimal size of a monotone DNF formula computing the k-CLIQUE function is Ω(n^{2/r}).

Rationale

The matroid rank captures the structural decomposition of the graph into disjoint cliques, and higher ranks imply that the graph can be decomposed into more cliques, which might allow for a more efficient DNF representation. Conversely, lower ranks (indicating fewer decompositions) would require larger DNFs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
ed': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
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TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
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TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
RESULT: SUPPORTED mean=40.00 std=0 support_fraction=1
```


Judge reasoning

Single instance tested with n=1 makes Ω(n^{2/r}) bound vacuous. Support fraction (1) meets threshold but insufficient data points violate pre-registered criterion. | next: Test with multiple instances across varying n and r values to validate scaling behavior

---

Matroid Rank Submodularity in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory (Polymatroid Expansion) × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 07:54 UTC
- **Entry ID**: `16b4e4e9c316`

Statement

Let μ(F) be the rank of the transversal matroid induced by a monotone DNF formula F. For all F with size poly(n), μ(F) = O(log n). For k-CLIQUE instances, μ(F) = Ω(n^{1 - 1/k})

Rationale

Transversal matroid rank captures submodularity of DNF terms, aligning with monotone circuit structure. k-CLIQUE's combinatorial constraints force high rank due to clique edge intersections, creating a gap between general DNFs and clique-specific formulas.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7104728.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7104728.py", line 63, in run_trial
    dnf = generate_dnf(n)
          ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7104728.py", line 24, in generate_dnf
    dnf.add(tuple(sorted(term) for term in terms))
TypeError: unhashable type: 'list'
```

Judge reasoning

Test crashed due to unhashable type error, preventing data collection to evaluate conjecture | next: Fix generate_dnf() to handle hashable types and re-run experiments

---

Tensor Product Multiplicity Inverse Proportional to Monotone Circuit Size for Permanent

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups (Tensor Product Decompositions) × Monotone Circuit Size for Permanent
- **Recorded**: 2026-05-12 08:01 UTC
- **Entry ID**: `7a5d68832c2b`

Statement

For a random 3-regular hypergraph H_n on n vertices, let m(λ) denote the multiplicity of irreducible representation λ in the decomposition of V_{n} ⊗ V_{n} under the symmetric group S_n. Then m(λ) = Θ(1/size_{mono}(Perm_n)) where size_{mono}(Perm_n) is the minimal monotone circuit size computing the permanent on n variables.

Rationale

Tensor product multiplicities encode combinatorial symmetries of permutation matrices, which are structurally similar to the monotone circuits required to compute permanent. The inverse proportionality suggests that richer representation structures correlate with lower circuit complexity, exposing hidden algebraic constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efe16db4.py", line 61, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efe16db4.py", line 40, in run_trial
    mλ = hook_length_formula(n, λ)
         ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efe16db4.py", line 30, in hook_length_f
   hook_lengths = [[n - row - col + 1 for col in range(col, n)] for row in range(row, n)]
                                                                                  ^^^
NameError: name 'row' is not defined. Did you mean: 'pow'?
```

Judge reasoning

Test crashed due to NameError in hook_length_formula implementation, preventing data collection | next: Fix the hook_length_formula implementation to properly define row/col indices and re-run trials

---

Persistent Homology of Read-Twice BPs Bounded by Log-Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent Homology × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 08:14 UTC
- **Entry ID**: `b4f04707e5bb`

Statement

For a read-twice BP P computing a function f: {0,1}^n → {0,1}, let β(P) = ∑_{k} (dim H_k) * persistence_k, where H_k is the k-th homology group of P's computation graph. Then β(P) = O(log size(P)), but β(IP_2) = Ω(n^2) where IP_2 is the inner product mod 2 function.

Rationale

Persistent homology captures hierarchical topological features of computation graphs. Read-once BPs have acyclic structures limiting homology complexity, while read-twice BPs may encode richer topological patterns. IP_2's exponential BP complexity creates persistent high-dimensional holes.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `1.05s`


```
c_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
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TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
```

```
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
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TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
RESULT: SUPPORTED mean=1.0 std=0.0 support_fraction=1.0
```

Judge reasoning

All 10 trials used the same single instance, violating the n-too-small failure mode. The support_fraction=1.0 is statistically vacuous for scaling claims. | next: Test with ≥80 diverse instances across varying n to validate scaling behavior

---

Hilbert Function Growth Inversely Proportional to Extended Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Hilbert Functions) × Extended Frege Proof Length
- **Recorded**: 2026-05-12 08:22 UTC
- **Entry ID**: `f78ced0ca719`

Statement

For a CNF tautology Φ with n variables, the Hilbert function H(ideal(Φ), d) at degree d = ⌈n/2⌉ satisfies H(ideal(Φ), d) = Θ(2^{-λ(n)}), where λ(n) is the Extended Frege proof length of Φ.

Rationale

Hilbert functions capture the complexity of polynomial ideals, which may encode logical dependencies in proofs. Inverse proportionality suggests deeper algebraic structure in proof systems, potentially revealing connections between algebraic geometry and proof complexity barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_906170a2.py", line 190, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_906170a2.py", line 156, in run_trial
    ideal = groebner_basis(phi, 2**32 - 5)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_906170a2.py", line 113, in groebner_bas
   s = subtract_matrices(basis[i], multiply_matrices([[basis[j][k] * mod_inverse(basis[i][j], mod)]]
```

```
                                                              ^
NameError: name 'k' is not defined
```

Judge reasoning

Test crashed due to NameError, preventing data collection. Support condition cannot be evaluated without results. | next: Fix the NameError in matrix operations by defining variable 'k' and revalidate the test implementation

---

Block Design Discrepancy and Communication Complexity Lower Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Design Theory × Communication Complexity (Discrepancy of Set Disjointness)
- **Recorded**: 2026-05-12 08:32 UTC
- **Entry ID**: `8a6982e09855`

Statement

For a symmetric (v, k, λ) block design, the discrepancy of the communication matrix for the set disjointness function is Ω(√(λ v / k))

Rationale

Block designs impose regular structure on incidence relations, which may constrain the discrepancy of associated communication matrices, offering a combinatorial lens to bound communication complexity

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
etric_value': None, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'design_not_
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
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TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
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TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
RESULT: FALSIFIED counterexample="design_not_possible" first_failing_seed=11
```

Judge reasoning

No empirical tests were executed (instances_tested=0), and the counterexample is a logical contradiction rather than an empirical failure. | next: Implement tests with concrete symmetric block design parameters and measure discrepancy values directly

---

Second Eigenvalue Inverse Proportional to Resolution Proof Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Proof Length for Tseitin Formulas
- **Recorded**: 2026-05-12 08:44 UTC
- **Entry ID**: `67d14839d492`

Statement

For a random Tseitin formula derived from a d-regular expander graph with n nodes, the resolution proof length is $\Theta(1 / \lambda_2)$, where λ_2 is the second largest eigenvalue of the adjacency matrix. For all such instances with $\lambda_2 \leq 1/n$, the proof length exceeds n^{1.5}.

Rationale

Expander graphs' spectral properties govern their resistance to local reasoning. The second eigenvalue λ_2 quantifies the graph's expansion, and its inverse may correlate with the difficulty of deriving contradictions via resolution, as expanders require global interactions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 101, in run_trial
    lambda_2 = eigenvalues(G)[1]
               ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 59, in eigenvalues
    y = matrix_multiplication(A, x)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 24, in matrix_multip
    C[i][j] += A[i][k] * B[k][j]
                         ~~~~^^^
TypeError: 'float' object is not subscriptable
```

Judge reasoning

Test crashed due to type error in matrix multiplication, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix matrix multiplication to handle vectors vs matrices correctly and retest with validated input types

---

Real Radical Dimension Bounds SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-12 09:26 UTC
- **Entry ID**: `02160132106f`

Statement

For a Max-CUT instance with n variables, the SOS degree required to approximate the maximum cut is at least the dimension of the real radical of the ideal generated by the constraint polynomials. Formally, if I is the ideal generated by {x_i^2 - x_i | 1 ≤ i ≤ n} ∪ {∑_{(i,j)∈E} (1 - x_i - x_j + x_i x_j) - α}, then dim(√I) ≥ d implies SOS degree ≥ d.

Rationale

The real radical captures the semialgebraic structure of the feasible region for Max-CUT. Higher-dimensional varieties may require higher-degree SOS certificates to capture their geometry, linking algebraic complexity to proof complexity. This conjecture bridges real algebraic geometry's intrinsic dimensionality with SOS hierarchy's computational requirements.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
'counterexample': ''}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
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TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
RESULT: SUPPORTED mean=20.0 std=0.0 support_fraction=1.0
```

Judge reasoning

parse_fail | next: NONE